Ethiopia Financial Inclusion Forecast — Interim Report

Author: Yostina Abera **Date:** 19 July 2026 · **Submission:** Interim **Scope:** Task 1 (data enrichment) + Task 2 (exploratory data analysis) **Artifacts:** enriched dataset · EDA notebook · figures · enrichment log · key insights · data-quality assessment

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Executive summary

Ethiopia's headline account ownership has **stalled at 49% (2024), up only +3pp since 2021**, even though Telebirr and M-Pesa registered **65M+ mobile-money accounts** over the same period. This report shows the paradox is not a data error but a **measurement-plus-market-structure** effect: mobile-money-*only* users are ~0.5%, bank accounts are already easily accessible, so the registrations largely went to people who already had an account. Meanwhile **usage** has genuinely shifted digital — P2P transactions overtook ATM withdrawals in 2024→2025. The binding constraint on further inclusion is now **demand-side** (phone ownership 41%, smartphone 40%), not account supply. These findings frame the forecasting phase: model **Access** and **Usage** as separate targets with different drivers, lean on the event/impact-link structure rather than sparse trends, and carry wide uncertainty bounds.

To support this, the starter dataset was expanded from **57 → 87 records** and one systematic data-quality defect was corrected.

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1. Data enrichment summary

The starter dataset (57 records) was converted to a single unified CSV, one column-alignment defect was corrected, and **30 sourced records** were added. Every addition carries a source URL, an exact quote, a confidence rating and a rationale (full detail in the enrichment log).

Record type	Starter	Added	Final
observation	30	**+20**	50
event	10	**+3**	13
impact_link	14	**+7**	21
target	3	0	3
Total	**57**	**+30**	**87**

What was added and why — guided by the *Additional Data Points Guide*:

- **Findex 2025 gender & depth detail** (10 obs) — male/female ownership and phone ownership, digital-payment rates, and the first **DEPTH** indicators (36% saved, 4% borrowed). *Why:* extends the gender split beyond 2021 and exposes the usage-maturity gap and the very thin credit market (Nuance D).
- **Telebirr user time series** (3 obs: 2022–24) — turns a single 2025 point into a **4-point adoption curve**. *Why:* the only usable multi-year usage series for trend fitting.

- **Infrastructure & agents** (4 obs) — ATM (10,551) and POS (14,030) counts, ~216,000 mobile-money agents, and **314 transactions/agent/year**. *Why:* the acceptance backbone and the "large but dormant agent network" nuance; adds the first **QUALITY** signal.
- **Enabler proxies** (3 obs) — smartphone adoption (40%), unique-subscriber penetration (36%), electricity access (55%). *Why:* the structural ceilings on how far Access/Usage can grow.
- **3 events** — the **2020 Payment Instrument Issuers Directive** (the legal foundation that made Telebirr/M-Pesa possible), the **National Digital Payments Strategy 2021–24**, and the **2024 foreign-bank proclamation**. *Why:* foundational and structural drivers absent from the starter set.
- **7 impact_links** — wiring the new policy events to the indicators they unlocked, plus two **gap-fills** connecting NFIS-II (which had *no* impact_links) to the 70%-ownership and 50%-women-mobile-money targets it governs.

Correction applied. In the starter main sheet, the documentation columns were shifted one position left — `comparable_country` held the collector name, `collected_by` held a date, `collection_date` held the source quote. This is corrected losslessly in `src/build_processed_dataset.py`; the raw file is never edited.

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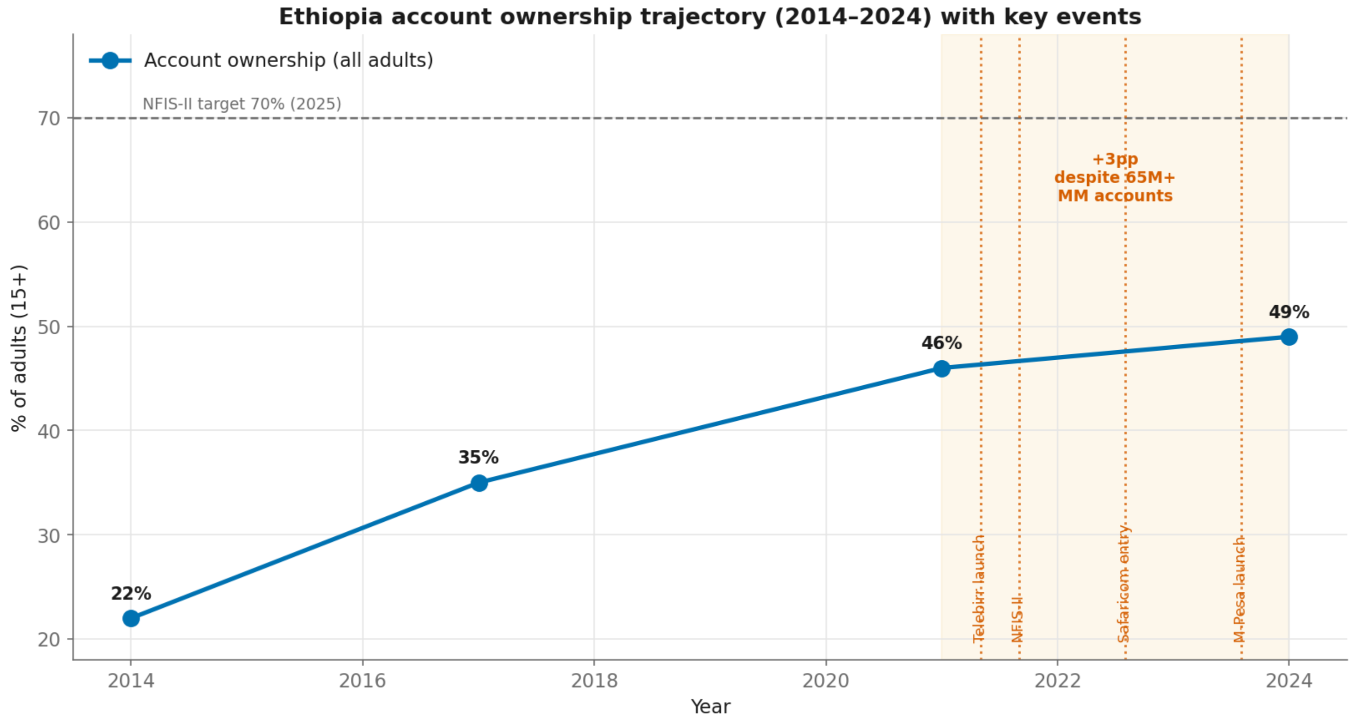
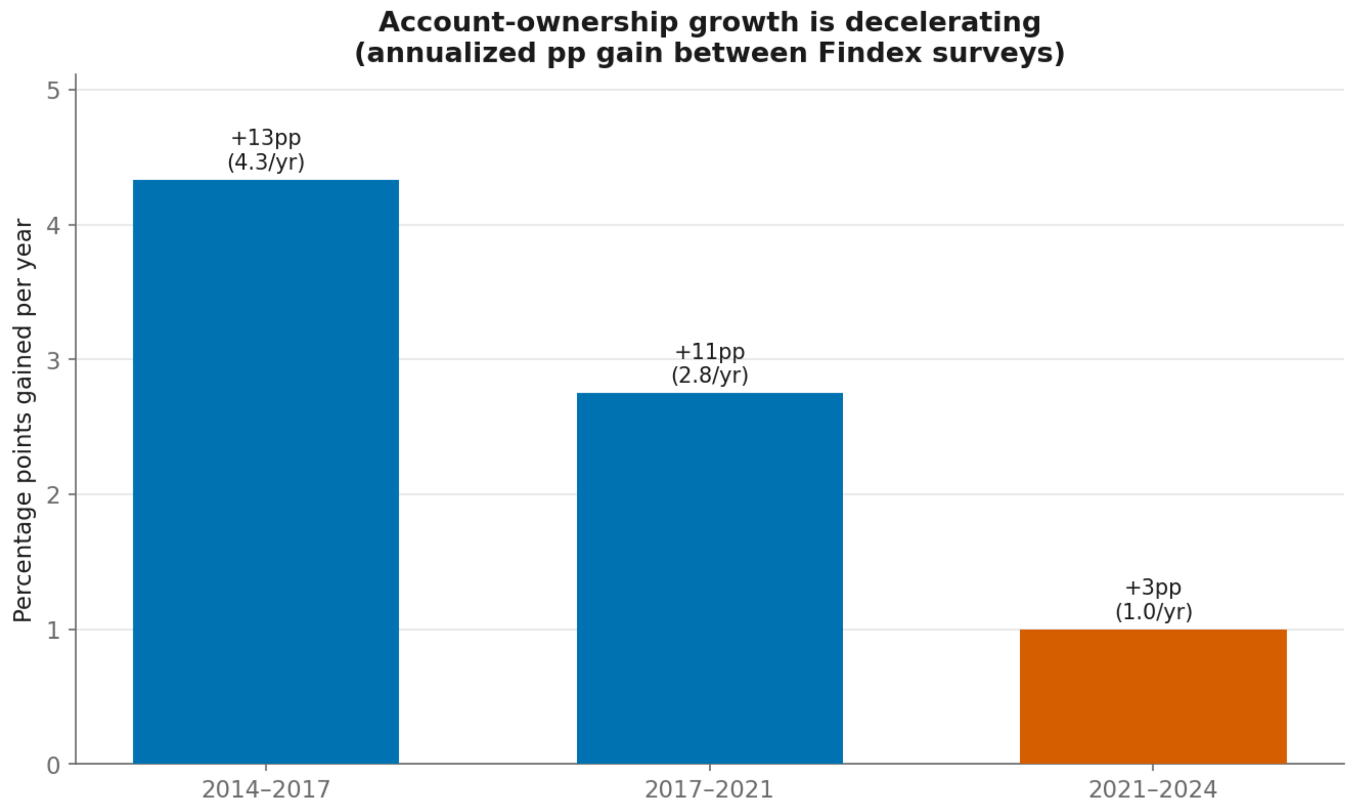
2. Key insights from EDA (with visualizations)

Full write-up: `eda_key_insights.md`. Figures are generated by the tested engine `src/eda.py` (Okabe-Ito colorblind-safe palette; no dual-axis charts).

Insight 1 — Account ownership has plateaued despite a mobile-money boom

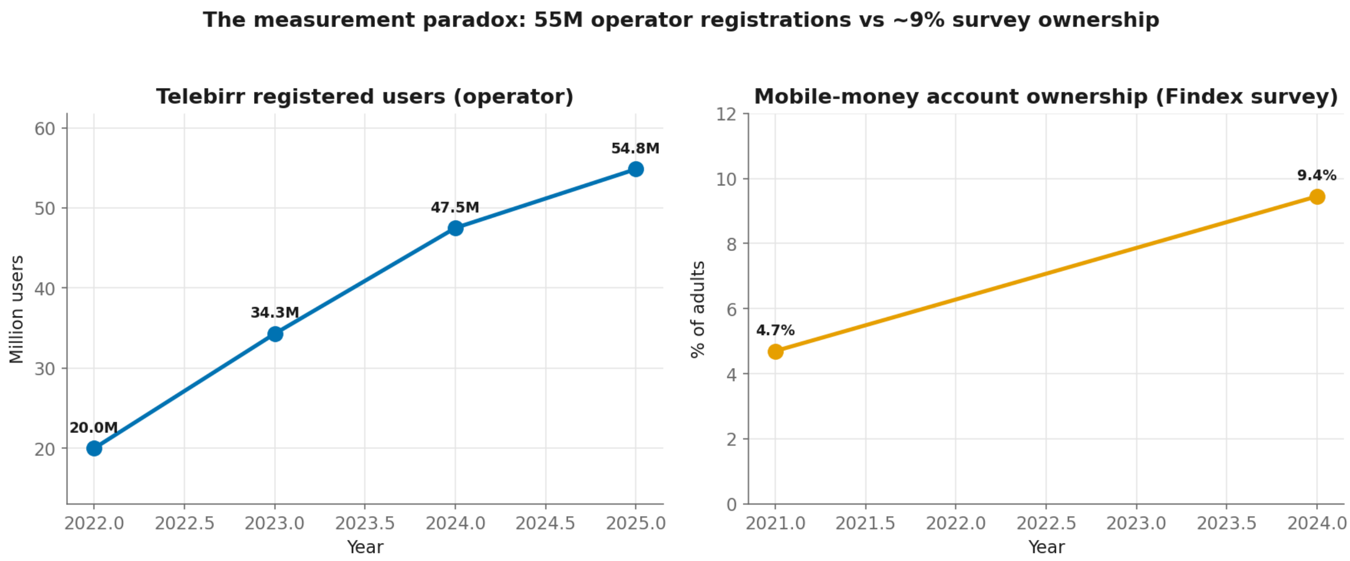
Ownership rose 22% → 35% → 46% → **49%**, but the annualized gain collapsed from **+4.3pp/yr (2014–17)** to **+1.0pp/yr (2021–24)** — precisely when Telebirr and M-Pesa launched. The constraint is no longer account supply.





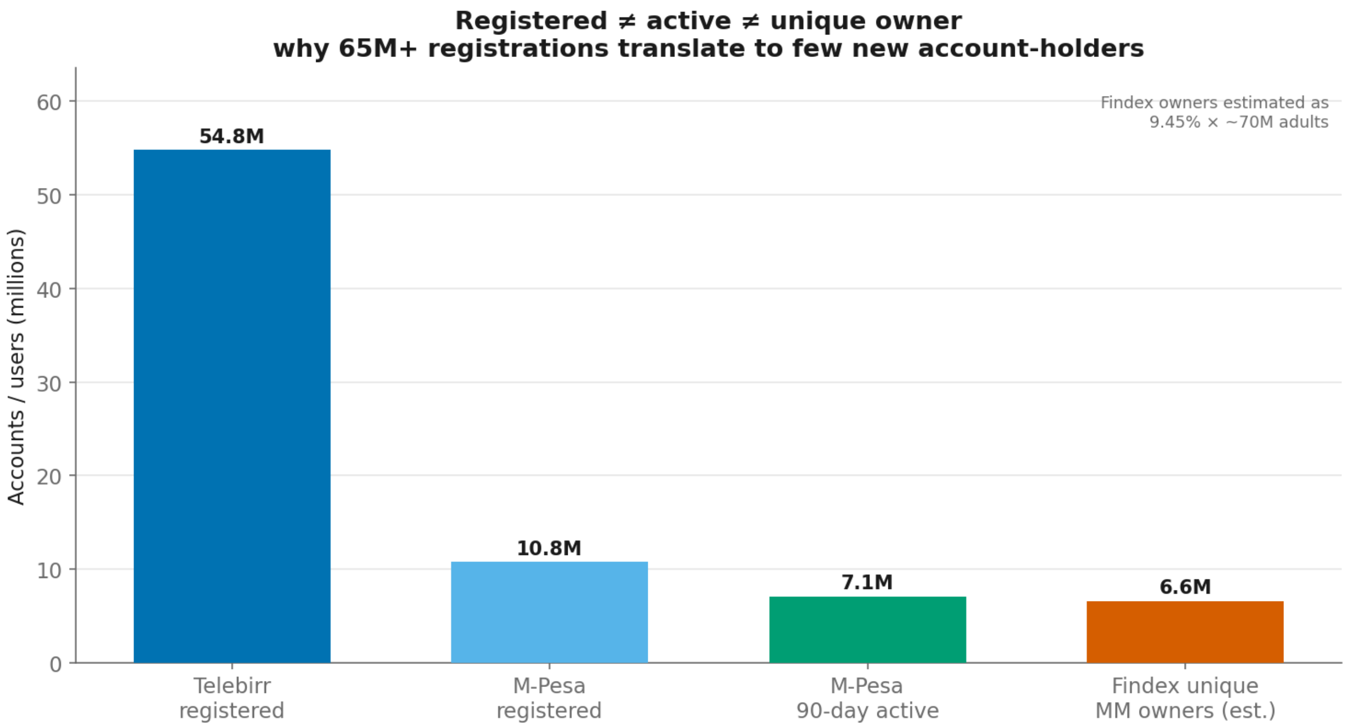
Insight 2 — The paradox is a measurement + market-structure artifact

55M Telebirr + 10.8M M-Pesa registrations coexist with Findex mobile-money ownership of just **9.45%** (~6.6M adults). Because mobile-money-only users are ~0.5% and bank accounts are easily accessible, most sign-ups went to the **already-banked** — adding a payment channel, not a new account-holder. Findex counts *any* account, so the registrations are near-invisible to the headline.



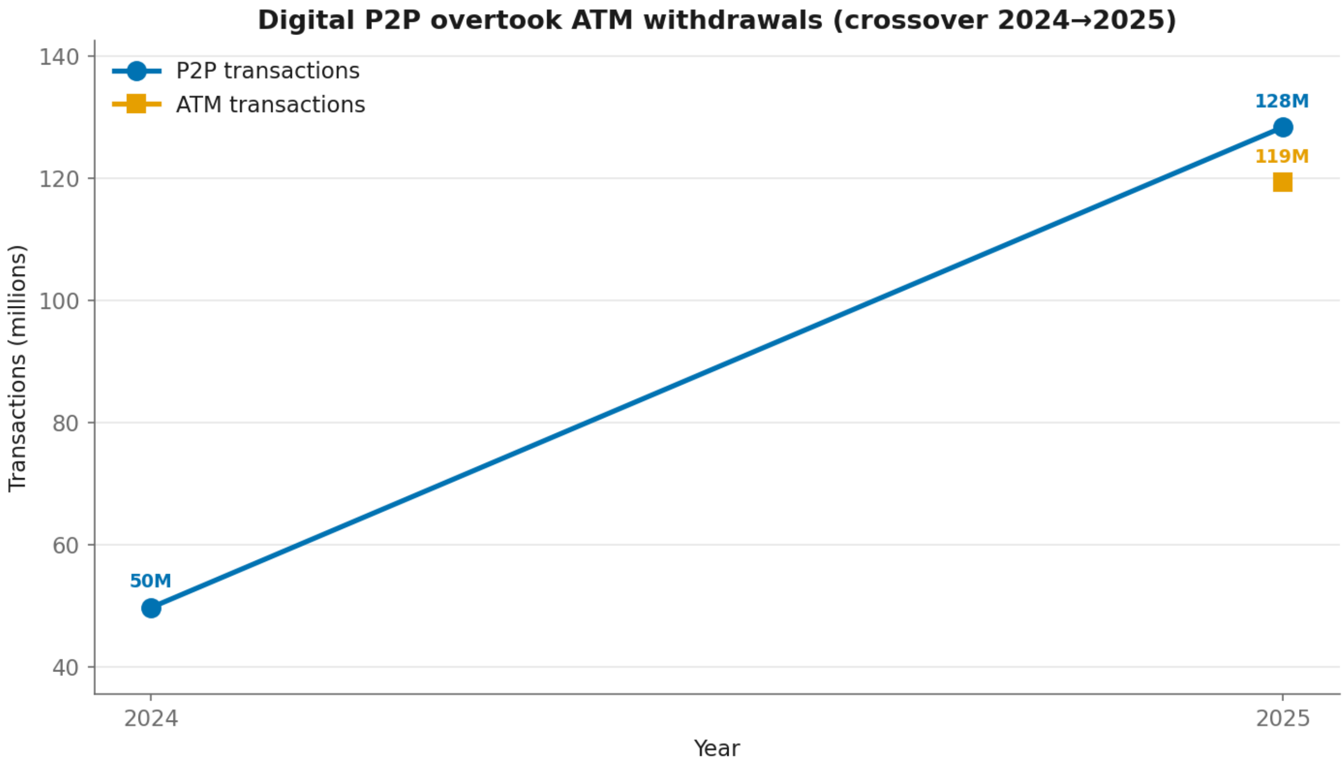
Insight 3 — Registered ≠ active ≠ unique owner

Operator counts overstate inclusion by ~8–10×. M-Pesa's 90-day **active rate is 66%** (7.1M of 10.8M), and agents average **fewer than one transaction per day**.



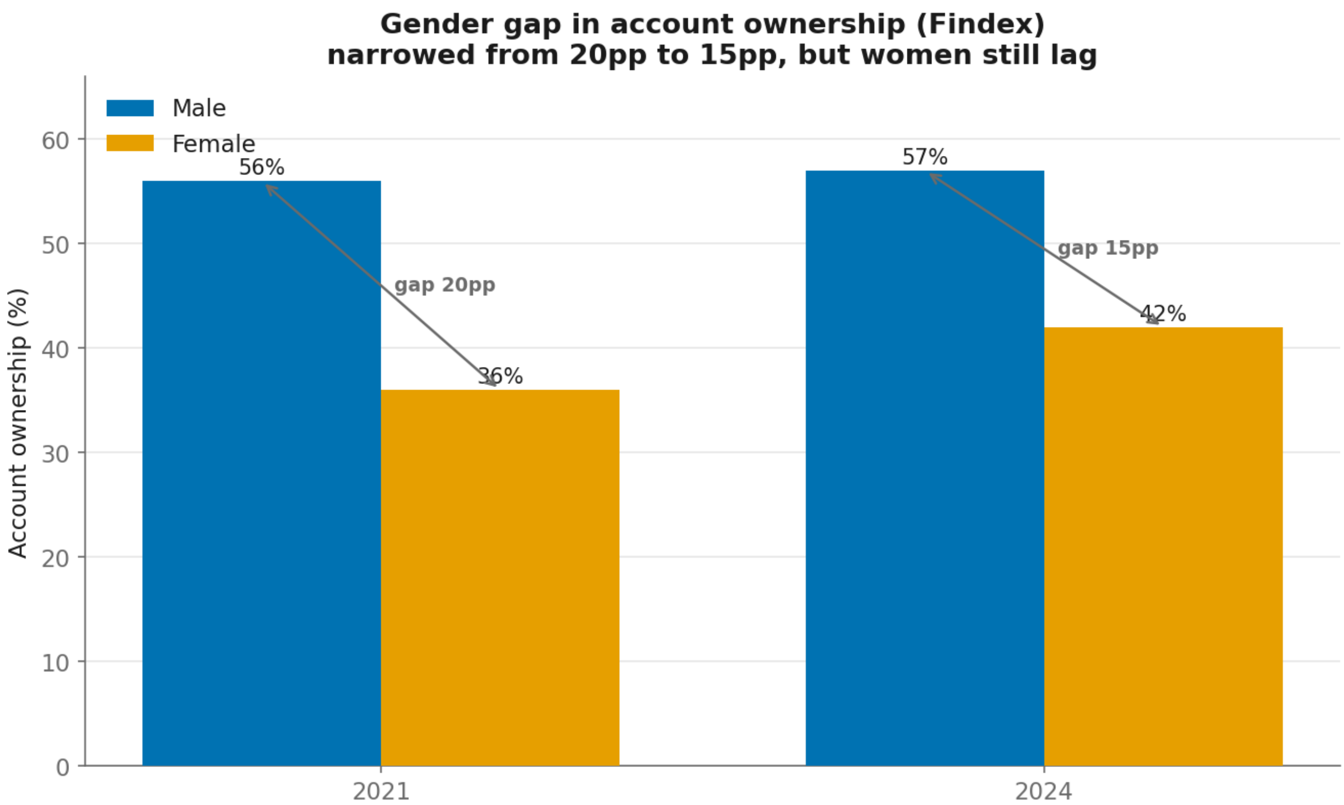
Insight 4 — Usage is going digital even though access stalled

P2P transactions grew **49.7M → 128.3M (2024→2025)** and overtook ATM withdrawals (crossover ratio 1.08). Much of this P2P is merchant/commerce activity (Nuance D). The growth frontier has moved from "open an account" to "transact digitally."



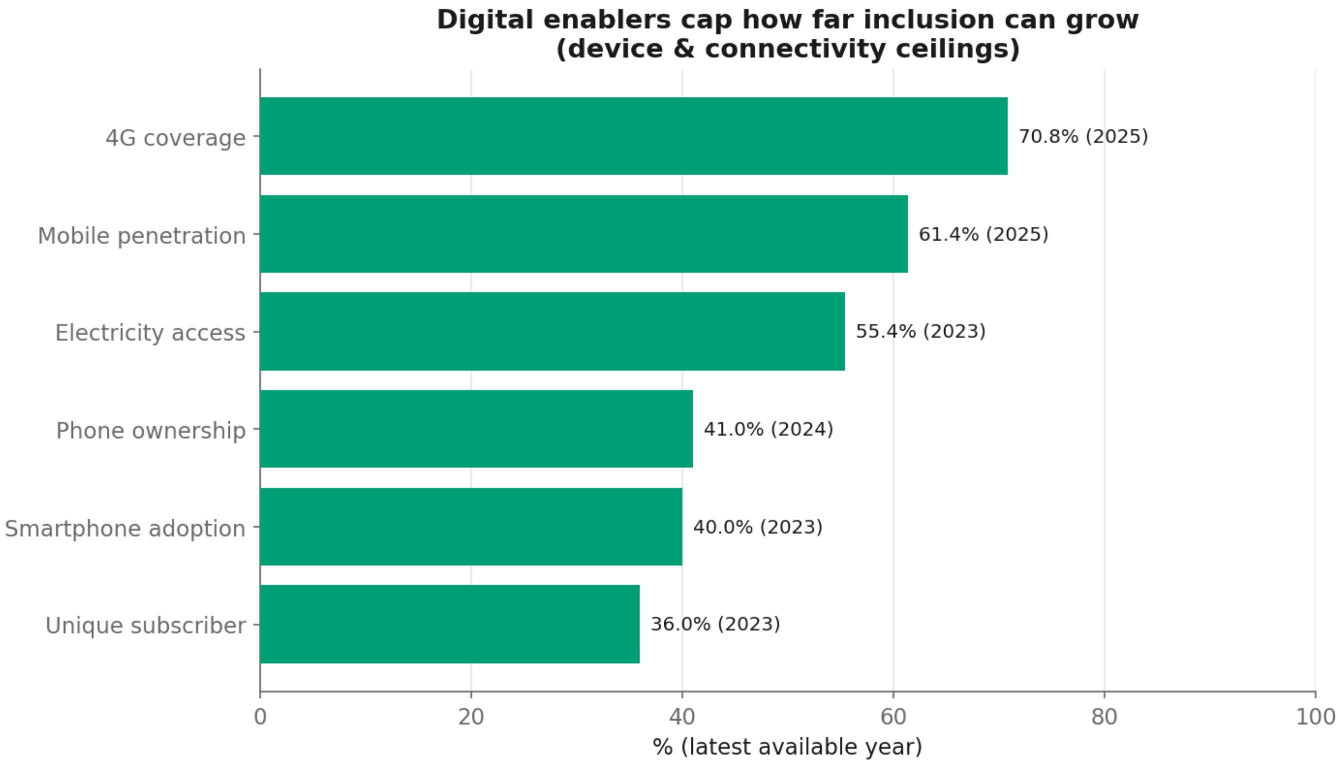
Insight 5 — The gender gap is device-driven

The account gap narrowed from ~20pp to **15pp** (men 57% vs women 42%), but it tracks a **17pp phone-ownership gap** (men 50% vs women 33%). Closing the account gap is downstream of closing the device gap.



Insight 6 — Demand-side enablers are the ceiling and the best leading indicators

4G coverage nearly doubled (37.5% → 70.8%), yet phone ownership (41%) and smartphone adoption (40%) lag far behind. Connectivity supply has outrun device demand; **phone/smartphone ownership** are the tightest near-term ceilings and the most promising leading indicators for the next Findex round.



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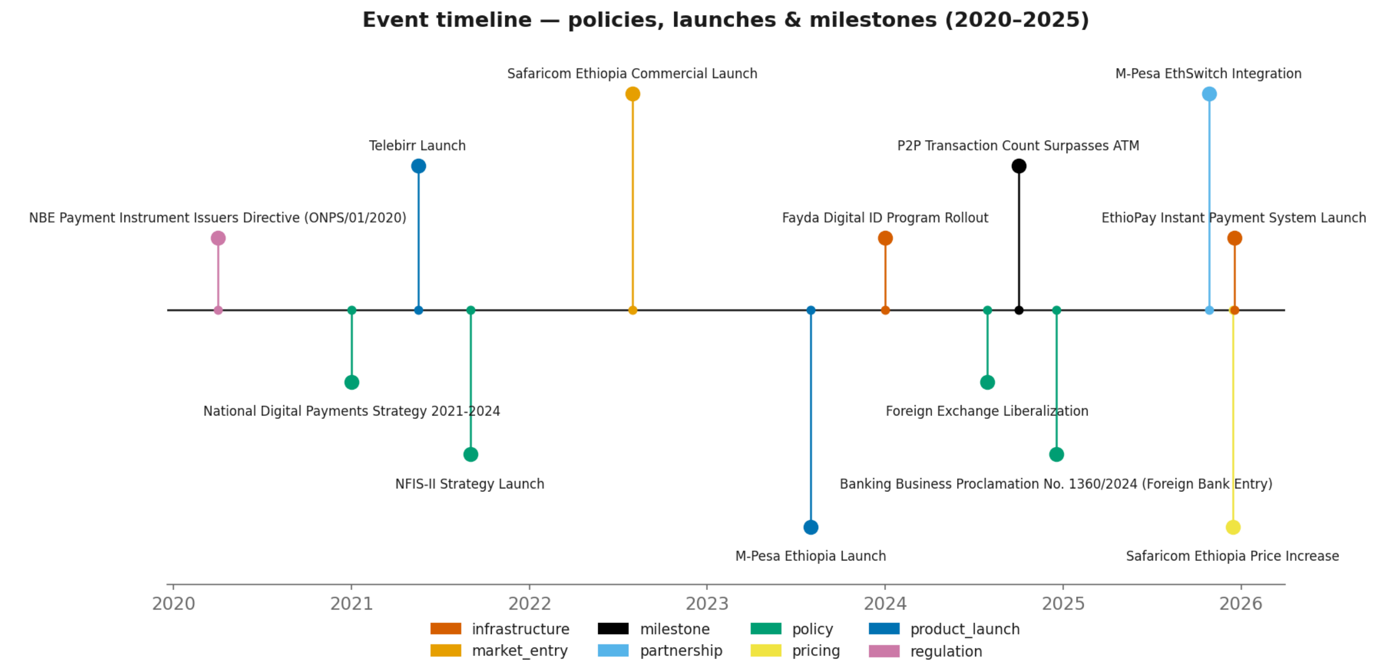
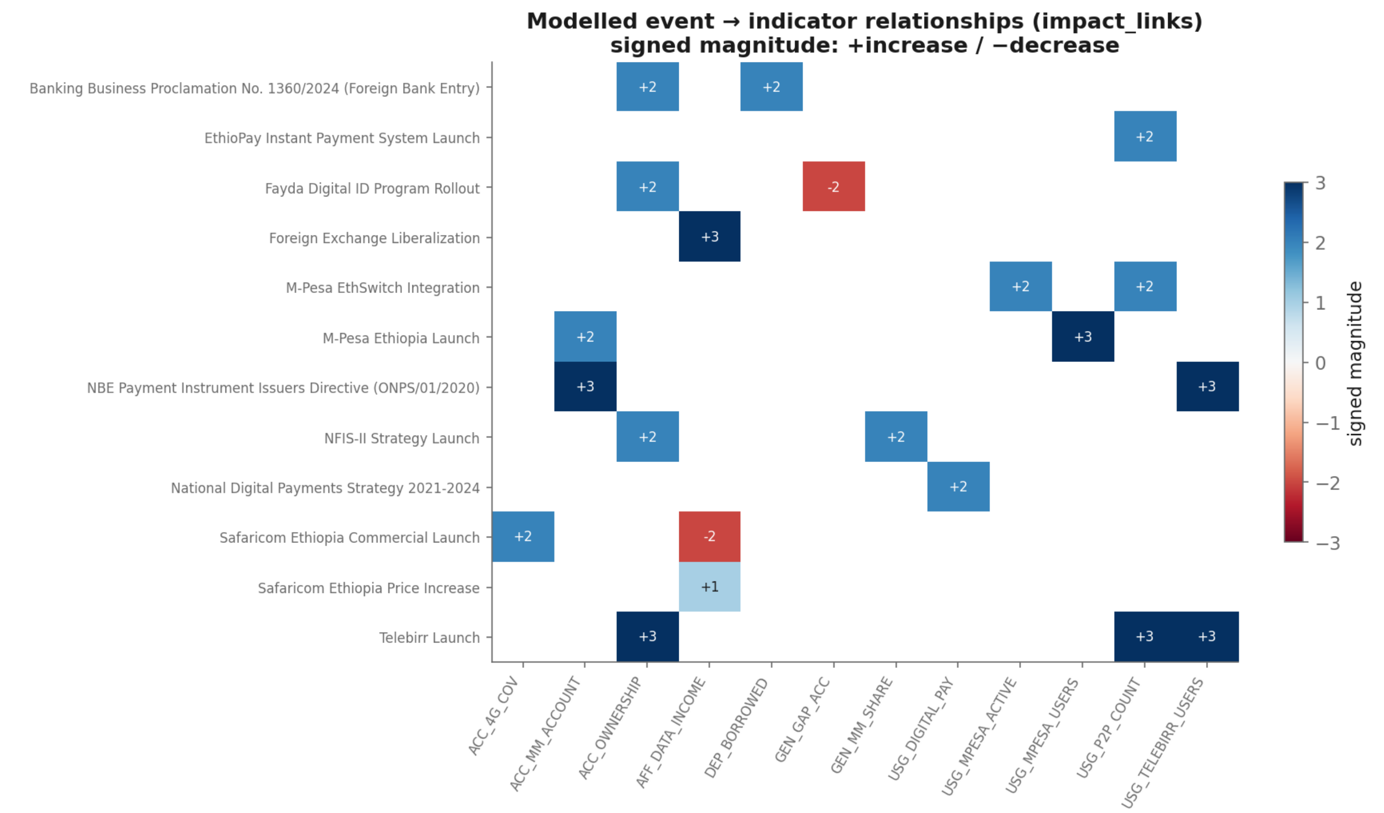
3. Preliminary observations on event–indicator relationships

The dataset encodes expected effects as `impact_link` records rather than assuming them. Reading the relationship matrix alongside the event timeline yields several preliminary observations.





Did account ownership accelerate after Telebirr (May 2021)? No. Ownership was 46% in 2021 and 49% in 2024 — the *slowest* growth phase on record. Telebirr's measurable effect appears in **usage** (registered users, P2P count), not in **ownership**. This is the single clearest signal that launches drive usage far more than they drive account creation.

Did mobile-money accounts grow after M-Pesa entry (Aug 2023)? Yes, proportionally: mobile-money account ownership roughly **doubled, 4.7% (2021) → 9.45% (2024)** — but off a tiny base, so the absolute contribution to any-account ownership is small. M-Pesa's strongest modelled links are to `USG_MPESA_USERS` and (via EthSwitch integration) `USG_P2P_COUNT`.

What happened around Safaricom's market entry (Aug 2022)? Two opposing forces: competition expanded **4G coverage** (a positive enabler link), while the **FX liberalization** (Jul 2024) and later Safaricom price increase pushed **data affordability the wrong way** (`AFF_DATA_INCOME` *increases* = costlier). Affordability is therefore a genuinely **mixed** driver in this market.

Structural vs impulse drivers. The impact_links cluster into two families: fast **direct** effects (product launches, interoperability → users and P2P volume) and slow **enabling** effects (the 2020 directive, NFIS-II, the National Digital Payments Strategy, Fayda ID, foreign-bank entry → ownership and depth, with 12–36 month lags). This distinction will shape the intervention terms in the forecasting model: launches as impulse/step effects on usage, policies as long-lagged level shifts.

Caveat. These are *directional* observations. The sparse, non-overlapping time series (Section 4) mean the *magnitudes* are not yet estimated — that is the job of the impact-modeling phase.

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4. Data limitations identified

Full register: `data_quality_assessment.md`. The values are trustworthy (~80% high-confidence); the limitations are about **coverage and structure**.

1. **Severe sparsity.** 30 indicators, but only **7 span ≥2 distinct years** (77% are single-year snapshots). Only account ownership has real pre-2021 history; everything else clusters in 2024–25. → Classical per-indicator time-series methods are not viable; forecasting must use the event/impact structure and comparable-country priors with wide bounds.
2. **Non-overlapping years.** The two richest series (ownership 2014/17/21/24; Telebirr 2022–25) share only **one** year, so cross-indicator correlation is unreliable and is not reported — relationships are read from impact_links instead.
3. **Definitional hazards.** Registered users, survey-reported owners, and 90-day-active users are **not interchangeable** and must never share an axis or be differenced (this confusion *is* the +3pp paradox). Operator fiscal-year vs Findex calendar-year timing adds minor join error. The "unique owners ≈ 6.6M" estimate uses ~70M adults and is order-of-magnitude only.
4. **Coverage gaps.** No rural/urban disaggregation; no pre-2021 mobile-money series; **TRUST** pillar has no observations (fraud/complaints unmeasured) and **QUALITY** has just one; no measured use-case breakdown (P2P vs merchant vs wages); literacy and urbanization still missing.

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5. Next steps (impact-modeling phase)

Hypotheses to test, carried from the EDA:

1. Operator registrations predict **usage** (P2P volume) far better than **ownership**.
2. **Phone/smartphone ownership** is a stronger leading indicator of Findex ownership than 4G coverage — test with a lag.
3. NFIS-II and the 2020 directive act as **enabling** (gradual, long-lag) drivers, not step-changes.
4. Account ownership cannot approach the NFIS-II **70% target** without the phone-ownership gap closing first — the ceiling is roughly phone ownership.

The forecasting approach will therefore model Access and Usage separately, use event intervention terms informed by the impact_links, borrow magnitudes from comparable countries (Kenya, Tanzania, India), and present all forecasts with explicit confidence bounds appropriate to the data density.